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22-23 April 2025

PUSH Trial #2 Genk

Autoneum. Mastering sound and heat.

PUSH HR for Genk

summary

- Due to technical issues, no good parts could be made during Trial #2
  - Iso and Polyol pumping set-up was not optimal. Most of the trial time was spent flushing the system. For this reason, only a few parts could be made and no process parameter could be investigated.
- In the few molded parts, foam remained unstable even after 3rd flushing
  - Potential root causes:
    - Flushing is insufficient
    - Isocyanate expired
    - Bad mixing of polyol prior to 1st flushing or before 3rd loading

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Testing of polyol prior to flushing

- Polyol and isocyanate taken from IBC after initial mixing of polyol
  Foam looks ok in terms of cell structure
- Slightly less rise height compared to sample made in Winterthur
  could be due to temperature difference of raw materials of 20°C (Genk) vs 25°C

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Testing of polyol during trial

Winterthur Genk

Winterthur Genk

- New polyol sample taken from day tank (TBC)
- Same procedure of mixing a total of 90g at MR0.60 and pouring blend into second cup
- At similar weight the rise height of the material from this trial remained lower
- The cell structure is very similar
- Very slight signs of more instability at the foam growing out of the top of the cup for the trial material

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parts

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- Strong flow of foam outside the cavity
- Reducing the shotweight reduced the flow-out, but collapses remained
- Skin looks ok, hence no real hint that RA is the root cause, but will still be tested in handmix tool in Winterthur

Comparison of blends in handmix tool

Tool: 40 x 40 x 2 cm

Mixer: 60 mm ring protected mixer at 2000 rpm, 5 s mixing time

Target part weight, 160g (density 50 g/L)

Samples:

Polyol:	Isocyanate
1. Blend made in Winterthur 16.04.2025 (reference)	1 L bottle provided by Dow in Oct 2025 (reference)
2. Blend from Genk IBC before flushing	sample from Genk IBC at start of trial
3. Blend from Genk IBC at end of trial	sample from Genk IBC at end of trial
4. Blend from Genk day tank at end of trial	sample from Genk day tank

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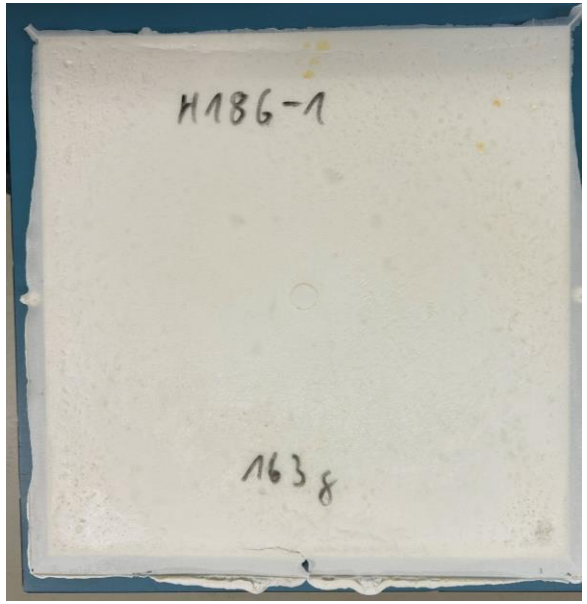
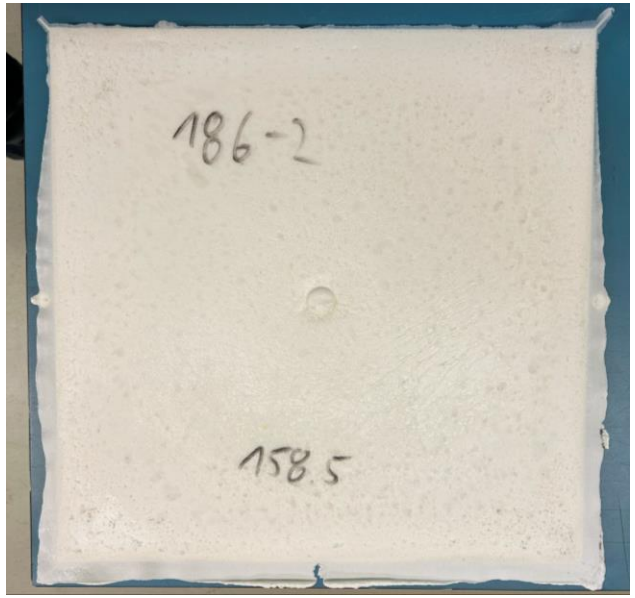
Comparison on reference and day tank samples

1 Ref polyol / ref isocyanate



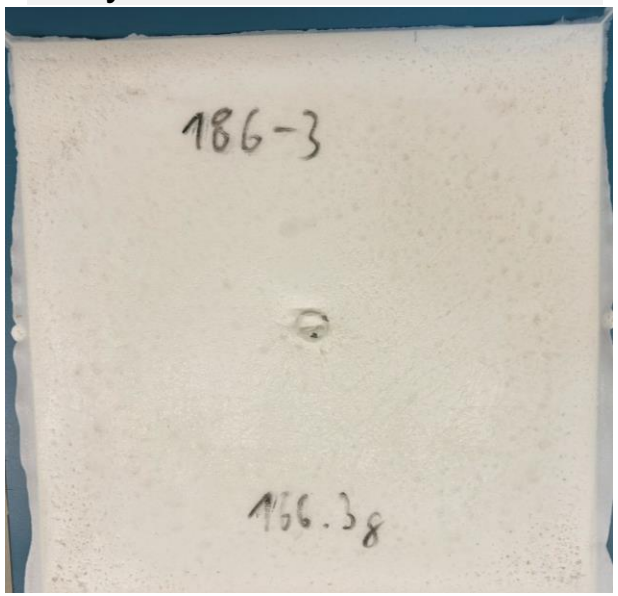
Center ok, voidy in corners

2 Day tank polyol / ref isocyanate



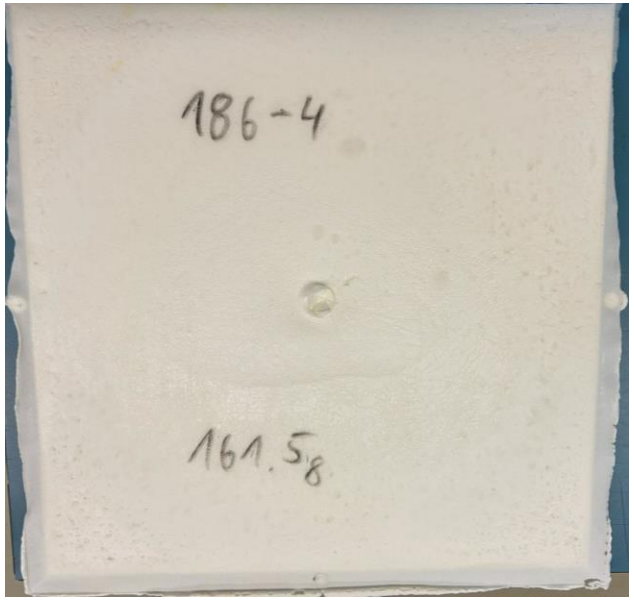
voidy in entire part

3 Day tank polyol / day tank isocyanate



voidy in entire part

4 Ref polyol / day tank isocyanate



Center ok, voidy in corners

The comparison shows that the polyol taken from the day tank caused instability in the foam sample. This can be caused by insufficient flushing or insufficient mixing of the IBC. The latter will be check in further tests.

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Cross sections of ref. vs. day tank samples

Foams made with reference polyol (1 and 4) show a finer and more even cell structure than samples made with the polyol taken from the day tank (2&3)

Next steps

- Further cross check in handmix trials of samples taken at Genk to further ensure the real root-cause
- Preparation for better flushing in next trial
- Order and prepare new polyol blend and fresh isocyanate for next trial
- Check potential improvements in HMI of dosing unit

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Flushing

The flushing was done in the following way:

- Drain the day tank.
- Load 100L of each component to the respective tank.
- Circulate the material several times over each of teh 4 mixing heads (30 minutes in total, switch MH after 60s)
- Drain day tanks and repeat another 2 times

In this way we only dilute the material, which is still in the ringline. Assuming we had 100L polyol in the ringline, we would still have 12.5% old material after the 3rd flushing. Filling up with 200L each we remain with a contamination of 4.2%

Action:

- Determine the volume in the ringline: length of ringline and inner diameter plus length & diameter of flexible hoses to and from mixing heads
- Make 3 way valve in return line in front of day tank workable (isocyanate side could not be switched) and connect a second drain pipe in addition to the one below the day tank